

KAOLACK, Senegal

WMO: 616790

Lat: 14.147N

Lon: 16.051W

Elev: 8

StdP: 101.23

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	18.2	-1.2	3.4	27.3	1.0	4.1	25.7	9.1	24.8	8.1	26.4	3.3	0	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.1	42.1	23.1	40.9	22.9	39.6	22.9	29.2	33.9	28.7	33.3	28.2	32.7	3.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	31.6	27.6	23.5	31.0	27.1	22.9	30.6	95.2	34.1	92.8	33.5	90.5	32.9	34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.8	6.2	13.7	44.5	2.0	0.8	12.2	45.1	11.0	45.6	9.9	46.1	8.4	46.7		
			8.9	30.5	1.3	1.3	8.0	31.4	7.2	32.2	6.5	32.9	5.6	33.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	26.6	28.3	30.2	31.1	31.2	30.8	29.8	29.0	29.0	30.5	29.7	27.3
	DBStd	2.27	2.03	2.15	2.28	2.12	1.81	1.40	1.47	1.45	1.45	1.53	1.42	1.71
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7104	513	513	628	633	658	623	614	588	571	636	592	536
	CDD18.3	4062	255	279	369	382	400	373	356	329	321	377	342	278
	CDH23.3	50068	2820	3404	4886	5124	5078	4642	4347	3798	3762	4862	4258	3087
	CDH26.7	27656	1521	2022	3134	3364	3107	2469	2039	1581	1660	2588	2473	1697
Wind	WSAvg	2.9	3.2	3.6	3.5	3.7	3.5	3.5	2.8	2.3	1.9	2.0	2.2	2.8
Precipitation	PrecAvg	612	1	1	0	0	2	39	139	220	165	54	4	1
	PrecMax	991	19	35	9	1	36	158	368	499	373	214	84	18
	PrecMin	309	0	0	0	0	0	0	7	16	11	1	0	0
	PrecStd	159	3	5	1	0	7	35	83	89	77	48	14	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	40.6	42.6	43.4	43.6	40.7	37.1	35.9	36.0	38.8	39.5	38.2
		MCWB	22.3	22.2	22.0	22.6	23.3	24.8	27.1	27.1	28.4	24.9	23.3	22.2
	2%	DB	36.6	39.0	41.2	42.1	41.9	38.4	35.7	34.2	34.7	37.3	38.3	37.0
		MCWB	21.6	21.8	22.2	22.9	23.7	25.5	27.0	27.4	28.2	25.9	23.2	21.8
	5%	DB	35.2	37.7	40.1	40.9	40.1	36.9	34.7	33.4	33.8	36.2	37.4	35.8
		MCWB	20.9	21.3	22.0	22.7	23.9	25.6	26.8	27.4	27.9	26.6	22.8	21.2
	10%	DB	33.9	36.1	38.5	39.3	38.2	35.5	33.7	32.4	32.9	35.2	36.3	34.4
		MCWB	20.1	20.5	21.7	22.7	24.1	25.6	26.7	27.2	27.6	26.8	22.6	20.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	26.8	26.6	27.8	28.2	29.0	29.5	29.4	29.7	29.8	28.1	26.1
		MCDB	34.8	36.1	38.6	38.4	38.0	35.3	33.8	33.0	33.9	34.5	34.8	34.7
	2%	WB	23.8	24.2	25.0	26.1	26.9	27.8	28.6	28.7	29.0	29.0	26.9	24.3
		MCDB	34.3	35.8	37.7	37.7	35.7	34.2	33.0	32.4	33.1	34.0	32.8	33.0
	5%	WB	22.4	22.9	23.8	24.8	26.1	27.1	27.9	28.1	28.5	28.4	26.0	23.1
		MCDB	33.0	34.9	36.3	36.3	34.6	33.4	32.3	31.9	32.4	33.1	31.7	32.7
	10%	WB	21.0	21.7	22.6	23.8	25.3	26.5	27.3	27.6	28.0	27.8	25.3	21.9
		MCDB	32.0	33.7	34.6	35.3	33.6	32.6	31.7	31.2	31.8	32.3	31.6	31.3
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.7	16.6	17.1	15.1	11.1	8.3	7.3	7.9	10.2	14.4	14.8
		MCDBR	16.7	17.4	18.2	18.3	17.6	13.0	9.6	8.6	8.9	11.6	15.7	16.8
		MCWBR	8.3	7.5	6.4	6.0	5.3	3.9	3.2	3.2	3.7	4.0	5.8	7.4
	5% WB	MCDBR	16.0	16.4	17.0	16.7	14.5	11.1	8.6	8.0	8.6	9.8	13.6	15.3
MCWBR		9.3	8.4	7.5	6.8	5.2	4.2	3.9	3.7	4.0	4.1	5.5	7.9	
Clear-Sky Solar Irradiance	taub	0.519	0.563	0.655	0.676	0.716	0.733	0.669	0.600	0.581	0.604	0.526	0.496	
	taud	1.870	1.769	1.590	1.607	1.537	1.584	1.727	1.890	1.922	1.846	1.944	1.969	
	Ebn at Noon	774	760	704	689	650	631	676	733	749	719	763	782	
	Edh at Noon	196	226	276	272	286	270	235	202	195	207	181	174	
All-Sky Solar Radiation	RadAvg	5.30	5.93	6.71	7.08	6.93	6.45	5.82	5.51	5.50	5.74	5.34	5.05	
	RadStd	0.26	0.30	0.25	0.23	0.25	0.16	0.21	0.22	0.25	0.23	0.25	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page